

EE 66
Fall 2026

Signals, Dynamics, and Information

Discussion 4B

1. Notes: Intro to DTFS

This is not a question, but a brief intro/overview of DTFS

A discrete-time signal $x : \mathbb{Z} \rightarrow \mathbb{C}$ is p -periodic, for an integer $p > 0$, if

$$x[n + p] = x[n] \quad \text{for all } n \in \mathbb{Z}.$$

The *fundamental period* is the smallest such p , and the *fundamental frequency* is $\omega_0 = 2\pi/p$.

Any p -periodic signal can be written as a linear combination of p complex exponentials, given by the *DTFS Synthesis Equation*:

$$x[n] = \sum_{k=0}^{p-1} X_k e^{ik\omega_0 n}, \quad \text{for all } n \in \mathbb{Z}.$$

The p coefficients X_k —called the *DTFS coefficients* of x —are unique.

The index k usually runs from 0 to $p - 1$. It may instead run over any set $\langle p \rangle$ of p consecutive integers, such as $[1, p]$ or $[-1, p - 2]$.

The DTFS coefficients describe the signal in the frequency domain. The magnitude $|X_k|$ tells you how much of the frequency $k\omega_0$ is present in x , and the phase of X_k gives the phase of that component. The coefficients are given by the *DTFS Analysis Equation*:

$$X_k = \frac{1}{p} \sum_{n=0}^{p-1} x[n] e^{-ik\omega_0 n}, \quad k = 0, 1, \dots, p - 1.$$

Answer: Cool, thanks!

2. Proving Cool DTFS Properties

Given a periodic signal x with period p and frequency $\omega_0 = \frac{2\pi}{p}$, recall the two golden equations for DTFS:

- Synthesis equation:

$$x[n] = \sum_{k \in \langle p \rangle} X_k e^{ik\omega_0 n}$$

- Analysis equation:

$$X_k = \frac{1}{p} \sum_{n \in \langle p \rangle} x[n] e^{-ik\omega_0 n}$$

Prove the following properties by "mathematically manipulating" these equations:

- (a) Prove that $X_k = X_{-k}^*$ if and only if x is a real-valued signal.

*Note: this is often called the **Conjugate Symmetry** property.*

Answer: We are proving an if and only if (sometimes denoted as "iff") claim, thus we must prove this from both directions!

We first prove the "if" direction (ie. the backwards direction): assume that x is a real-valued signal. We want to show that $X_k = X_{-k}^*$. We start from the analysis equation to find the conjugate of X_{-k} :

$$X_{-k} = \frac{1}{p} \sum_{n \in \langle p \rangle} x[n] e^{ik\omega_0 n}.$$

Conjugate both sides, distributing the conjugate over the sum:

$$X_{-k}^* = \frac{1}{p} \sum_{n \in \langle p \rangle} x^*[n] e^{-ik\omega_0 n}.$$

Since x is real-valued, $x^*[n] = x[n]$, so

$$X_{-k}^* = \frac{1}{p} \sum_{n \in \langle p \rangle} x[n] e^{-ik\omega_0 n} = X_k,$$

by the analysis equation. $\square$

Next, we prove the "only if" direction: assume that $X_k = X_{-k}^*$. We want to show that $x \in \mathbb{R}$:

Since we know something about X_k , it is easier to use the synthesis equation:

$$x[n] = \sum_{k \in \langle p \rangle} X_k e^{ik\omega_0 n}.$$

Conjugate both sides:

$$\begin{aligned} x^*[n] &= \sum_{k \in \langle p \rangle} X_k^* e^{-ik\omega_0 n} \\ &= \sum_{k \in \langle p \rangle} X_{-k} e^{-ik\omega_0 n} \\ &= \sum_{m \in \langle p \rangle} X_m e^{im\omega_0 n} \\ &= x[n]. \end{aligned}$$

The second line used the hypothesis $X_k = X_{-k}^*$, so $X_k^* = X_{-k}$; the third re-indexed with $m = -k$; and the last is the synthesis equation for x .

Hence $x[n] = x^*[n]$ for every n , so x is real-valued. $\square$

- (b) Let x be a discrete-time periodic signal with DTFS coefficients X_k . Let $y[n] = x[n - n_0]$ be a time-shifted version of x with DTFS coefficients Y_k . Prove that $Y_k = e^{-ik\omega_0 n_0} X_k$.

*Note: this is often called the **Time Shift** property.*

Answer: We want to show that in the DTFS expansion of y , the coefficient of $e^{ik\omega_0 n}$ is $e^{-ik\omega_0 n_0} X_k$. We know the coefficients of x , not those of y , so start from the synthesis equation for x :

$$x[n] = \sum_{k \in \langle p \rangle} X_k e^{ik\omega_0 n}.$$

Then, by definition of y ,

$$\begin{aligned} y[n] &= x[n - n_0] \\ &= \sum_{k \in \langle p \rangle} X_k e^{ik\omega_0 (n - n_0)} \\ &= \sum_{k \in \langle p \rangle} X_k e^{ik\omega_0 n} e^{-ik\omega_0 n_0} \\ &= \sum_{k \in \langle p \rangle} \left(e^{-ik\omega_0 n_0} X_k \right) e^{ik\omega_0 n}. \end{aligned}$$

But the synthesis equation for y is

$$y[n] = \sum_{k \in \langle p \rangle} Y_k e^{ik\omega_0 n}.$$

Since the DTFS representation of a periodic signal is unique, the coefficient multiplying $e^{ik\omega_0 n}$ must be the same in both expressions. Therefore,

$$Y_k = e^{-ik\omega_0 n_0} X_k.$$

The factor $e^{-ik\omega_0 n_0}$ is a phase shift of the DTFS coefficient X_k : a time shift in the signal produces a phase shift in its DTFS coefficients.

3. DTFS Analysis of a Simple Signal

Let $x[n]$ be the periodic signal with one period given by $x = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 1 \end{bmatrix}$, i.e. $x[0] = 2, x[1] = 1, x[2] = 0, x[3] = 1$, repeating with period 4.

- (a) Find the DTFS coefficients X_k for $k = 0, 1, 2, 3$. Then, using these coefficients, write $x[n]$ in the form $A + B \cos(\omega_0 n)$.

Answer: With $p = 4, \omega_0 = 2\pi/4 = \pi/2$. Using $X_k = \frac{1}{4} \sum_{n=0}^3 x[n] e^{-ik\omega_0 n}$:

$$X_0 = \frac{1}{4}(2 + 1 + 0 + 1) = 1,$$

$$X_1 = \frac{1}{4}(2 - i + 0 + i) = \frac{1}{2},$$

$$X_2 = \frac{1}{4}(2 - 1 + 0 - 1) = 0,$$

$$X_3 = \frac{1}{4}(2 + i + 0 - i) = \frac{1}{2}.$$

Notice that $X_3 = \overline{X_1}$ and X_0, X_2 are real, as expected from conjugate symmetry. Since $e^{i3\pi n/2} = e^{-i\pi n/2}$ for integer n , synthesis gives

$$x[n] = 1 + \frac{1}{2}e^{i\pi n/2} + \frac{1}{2}e^{-i\pi n/2} = 1 + \cos\left(\frac{\pi}{2}n\right).$$

This gives the samples $[2, 1, 0, 1]$ at $n = 0, 1, 2, 3$.

- (b) Let $y[n] = x[n - 1]$ be a time-shifted version of $x[n]$. Find the DTFS coefficients Y_k of $y[n]$.

Answer: By the time-shift property, $y[n] = x[n - N]$ has coefficients $Y_k = e^{-ik\omega_0 N} X_k$. Here $N = 1$ and $\omega_0 = \pi/2$, so

$$Y_0 = 1(1) = 1,$$

$$Y_1 = (-i)\left(\frac{1}{2}\right) = -\frac{i}{2},$$

$$Y_2 = (-1)(0) = 0,$$

$$Y_3 = i\left(\frac{1}{2}\right) = \frac{i}{2}.$$

As a check, shifting the samples gives $y = [1, 2, 1, 0]^\top$, and direct analysis gives

$$Y_1 = \frac{1}{4}(1 + 2(-i) + 1(-1) + 0(i)) = -\frac{i}{2},$$

agreeing with the time-shift calculation.